ภาคผนวก ก

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ASP	Active Server Page
CDC	Connect Device Configuration
CGI	Common Gateway Interface
CLDC	Connected, Limited Device Configuration
HTML	HyperText Markup Language
J2EE	Java 2 Enterprise Edition
J2ME	Java 2 Micro Edition
J2SE	Java 2 Standard Edition
JSP	Java Server Page
JVM	Java Virtual Machine
KVM	K Virtual Machine
MIDP	Mobile Information Device Profile
PDA	Personal Digital (Data) Assistant
PHP	PHP Hypertext Preprocessor
SMS	Short messaging service
WAP	Wireless Application Protocol

ภาคผนวก ข

ซอร์ซโค้ดของ applet (Map.jsp)

```
import java.awt.*;
import java.applet.*;
import java.awt.event.*;
import java.util.*;
import java.io.*;
import java.sql.*;
import java.text.*;

public class Map1 extends Applet implements MouseListener,MouseMotionListener
{
        private Image i;
        private int x,y,clickx,clicky,picx,picy,maxpicx,maxpicy,minpicx,minpicy;
        private int count = 0;
        private int x_coord[] = new int[100];
        private int y_coord[] = new int[100];
        String temp = "",getdistrict = "",getfirstroute = "";

        public int CountXY(String image) throws ClassNotFoundException,SQLException
        {
                int count_xy = 0;

                try
                {
                        Class.forName("org.gjt.mm.mysql.Driver");
                        Connection mycon =
DriverManager.getConnection("jdbc:mysql://localhost/myproject?user=root&password=231025231025");
                        Statement stmt = mycon.createStatement();
```

```
                    ResultSet rs = stmt.executeQuery("select count(*) as num from mapinfo where
image = '"+image+"' ");
                    while (rs.next())
                    {
                              count_xy = rs.getInt("num");
                    }

                    rs.close();
                    stmt.close();
                    mycon.close();
          }
          catch(Exception ex)
          {
                    count_xy = 0;
          }

          return count_xy;
     }

     public int[] getX(String image) throws ClassNotFoundException,SQLException
     {
          int x[];
          x = new int[100];
          int i = 0;

          try
          {
                    Class.forName("org.gjt.mm.mysql.Driver");
                    Connection mycon =
DriverManager.getConnection("jdbc:mysql://localhost/myproject?user=root&password=231025231025");
```

```
                        Statement stmt = mycon.createStatement();

                        ResultSet rs = stmt.executeQuery("select x_coord from mapinfo where image =
'"+image+"' ");
                        while (rs.next())
                        {
                                x[i] = rs.getInt("x_coord") - 5;
                                i++;
                        }

                        rs.close();
                        stmt.close();
                        mycon.close();
                }
                catch(Exception ex)
                {
                }

                return x;
        }

        public int[] getY(String image) throws ClassNotFoundException,SQLException
        {
                int y[];
                y = new int[100];
                int i = 0;

                try
                {
                        Class.forName("org.gjt.mm.mysql.Driver");
```

```
                    Connection mycon =
DriverManager.getConnection("jdbc:mysql://localhost/myproject?user=root&password=231025231025");
                    Statement stmt = mycon.createStatement();

                    ResultSet rs = stmt.executeQuery("select y_coord from mapinfo where image =
'"+image+"' ");
                    while (rs.next())
                    {
                        y[i] = rs.getInt("y_coord") - 5;
                        i++;
                    }

                    rs.close();
                    stmt.close();
                    mycon.close();
                }
                catch(Exception ex)
                {
                }

                return y;
        }

        public String shownode(String image,int x,int y) throws ClassNotFoundException,SQLException
        {
                String node = "";

                try
                {
                    Class.forName("org.gjt.mm.mysql.Driver");
```

```
                    Connection mycon =
DriverManager.getConnection("jdbc:mysql://localhost/myproject?user=root&password=231025231025");
                    Statement stmt = mycon.createStatement();

                    ResultSet rs1 = stmt.executeQuery("select node from mapinfo where image =
'"+image+"'  && x_coord > "+(x-5)+" && x_coord < "+(x+5)+" && y_coord > "+(y-5)+" && y_coord <
"+(y+5)+"");
                    while (rs1.next())
                    {
                              node = rs1.getString("node");
                    }
                    rs1.close();

                    stmt.close();
                    mycon.close();
          }
          catch(Exception ex)
          {
          }

          return node;
     }

     public String getPic(String district) throws ClassNotFoundException,SQLException
     {
          try
          {
                    Class.forName("org.gjt.mm.mysql.Driver");
                    Connection mycon =
DriverManager.getConnection("jdbc:mysql://localhost/myproject?user=root&password=231025231025");
                    Statement stmt = mycon.createStatement();
```

```
                        ResultSet rs = stmt.executeQuery("select image from mapinfo where
default_district = '"+district+"' ");
                        while (rs.next())
                        {
                                temp = rs.getString("image");
                        }

                        rs.close();
                        stmt.close();
                        mycon.close();
                }
                catch (Exception ex)
                {
                }

                if (temp.equals(""))
                {
                        temp = "e10047000n1346000s4.jpg";
                }

                return temp;
        }

        public String showdetail(String node)  throws ClassNotFoundException,SQLException
        {
                String detail = "";

                try
                {
                        Class.forName("org.gjt.mm.mysql.Driver");
```

```
                              Connection mycon =
DriverManager.getConnection("jdbc:mysql://localhost/myproject?user=root&password=231025231025");
                              Statement stmt = mycon.createStatement();

                              ResultSet rs1 = stmt.executeQuery("select * from nodeadjacent where node =
'"+node+"' ");
                              while (rs1.next())
                              {
                                        detail = detail + " GOTO " + rs1.getString("adja") + "(" +
rs1.getInt("distance") + "," + rs1.getInt("expressway") + "," + rs1.getInt("density") + ")";
                              }
                              rs1.close();

                              stmt.close();
                              mycon.close();
                    }
                    catch(Exception ex)
                    {
                    }

                    return detail;
          }

          public void init()
          {
                    //picx = 10047000;
                    //picy = 1346000;
                    String temp = "";
                    getdistrict = getParameter("map_district");
                    getfirstroute = getParameter("map_image");
                    try
```

```
		{
			if ((getfirstroute.equals("")) || (getfirstroute == null))
			{
				temp = getPic(getdistrict);
				picx = Integer.parseInt(temp.substring(1,9));
				picy = Integer.parseInt(temp.substring(10,17));
			}
			else
			{
				picx = Integer.parseInt(getfirstroute.substring(1,9));
				picy = Integer.parseInt(getfirstroute.substring(10,17));
			}
		}
		catch(Exception ex)
		{
			picx = 10047000;
			picy = 1346000;
		}
		maxpicx = 10047000;
		maxpicy = 1346000;
		minpicx = 10044600;
		minpicy = 1343600;

		i  =  getImage (getCodeBase(),"pic/e"+picx+"n"+picy+"s4.jpg");
		addMouseListener(this);
		addMouseMotionListener(this);
	}

	public void paint (Graphics g)
	{
		try
```

```
        {
            count = CountXY("e"+picx+"n"+picy+"s4");
            x_coord = getX("e"+picx+"n"+picy+"s4");
            y_coord = getY("e"+picx+"n"+picy+"s4");
        }
        catch (Exception e)
        {
            count = 0;
        }

        g.drawImage(i,0,0,this);
        //g.drawString("|"+getfirstroute+"|",10,10);
        //g.drawString("e"+picx+"n"+picy+"s4",10,50);
        //g.drawString("|"+getdistrict+"|",10,80);
        //String temp1 = Integer.toString(count);
        //g.drawString(temp1,10,10);
        //String temp2 = Integer.toString(x[0]);
        //g.drawString(temp2,10,70);
        //String temp3 = Integer.toString(y[0]);
        //g.drawString(temp3,10,70);

        for (int i=0;i<count;i++)
        {
            g.fillOval(x_coord[i],y_coord[i],10,10);
        }
    }

    public void mousePressed(MouseEvent e) {}
    public void mouseReleased(MouseEvent e) {}
    public void mouseExited(MouseEvent e)
    {
```

```
        showStatus("Pointer outside applet");
    }
    public void mouseEntered(MouseEvent e) {}
    public void mouseClicked(MouseEvent e)
    {
        clickx = e.getX();
        clicky = e.getY();
        showStatus("clicked at "+clickx+","+clicky);
        //move left
        if ((clickx < 20) && (clicky>290) && (clicky<310))
        {
            if (picx == minpicx)
            {
                showStatus("Can't move left");
            }
            else
            {
                showStatus("click move left");
                MoveLeft();
            }
        }
        //move right
        if ((clickx >580) && (clicky>290) && (clicky<310))
        {
            if (picx == maxpicx)
            {
                showStatus("Can't move right");
            }
            else
            {
                showStatus("click move right");
```

```
				MoveRight();
			}
		}
		//move up
		if ((x < 310) && (x>290) && (y<20))
		{
			if (picy == maxpicy)
			{
				showStatus("Can't move up");
			}
			else
			{
				showStatus("click move up");
				MoveUp();
			}
		}
		//move down
		if ((x < 310) && (x>290) && (y>580))
		{
			if (picy == minpicy)
			{
				showStatus("Can't move down");
			}
			else
			{
				showStatus("click move down");
				MoveDown();
			}
		}
		//move up left
		if ((x <20) && (y<20))
```

```
{
        if ((picx == minpicx) || (picy == maxpicy))
        {
                showStatus("Can't move up left");
        }
        else
        {
                showStatus("click move up left");
                MoveUpLeft();
        }
}
//move down left
if ((x < 20) && (y>580))
{
        if ((picx == minpicx) || (picy == minpicy))
        {
                showStatus("Can't move down left");
        }
        else
        {
                showStatus("click move down left");
                MoveDownLeft();
        }
}
//move up right
if ((x > 580) && (y<20))
{
        if ((picx == maxpicx) || (picy == maxpicy))
        {
                showStatus("Can't move up right");
        }
```

```
				else
				{
					showStatus("click move up right");
					MoveUpRight();
				}
			}
			//move down right
			if ((x >580) && (y>580))
			{
				if ((picx == maxpicx) || (picy == minpicy))
				{
					showStatus("Can't move down left");
				}
				else
				{
					showStatus("click move down right");
					MoveDownRight();
				}
			}
		}

		public void mouseDragged(MouseEvent e) {}
		public void mouseMoved(MouseEvent e)
		{
			x = e.getX();
			y = e.getY();
			showStatus(x+","+y);
			try
			{
				showStatus(shownode("e"+picx+"n"+picy+"s4",x,y) +
showdetail(shownode("e"+picx+"n"+picy+"s4",x,y)));
```

```
}
catch(Exception ex)
{
}
if ((x < 20) && (y>290) && (y<310))
{
        showStatus("Move left");
}
if ((x >580) && (y>290) && (y<310))
{
        showStatus("Move right");
}
if ((x < 310) && (x>290) && (y<20))
{
        showStatus("Move up");
}
if ((x < 310) && (x>290) && (y>580))
{
        showStatus("Move down");
}
if ((x <20) && (y<20))
{
        showStatus("Move up left");
}
if ((x < 20) && (y>580))
{
        showStatus("Move down left");
}
if ((x > 580) && (y<20))
{
        showStatus("Move up right");
```

```
        }
        if ((x >580) && (y>580))
        {
                showStatus("Move down right");
        }
}

public void MoveLeft()
{
        picx = picx - 2400;
        i = getImage (getCodeBase(),"pic/e"+picx+"n"+picy+"s4.jpg");
        repaint();
}
public void MoveRight()
{
        picx = picx + 2400;
        i = getImage (getCodeBase(),"pic/e"+picx+"n"+picy+"s4.jpg");
        repaint();
}
public void MoveUp()
{
        picy = picy + 2400;
        i = getImage (getCodeBase(),"pic/e"+picx+"n"+picy+"s4.jpg");
        repaint();
}
public void MoveDown()
{
        picy = picy - 2400;
        i = getImage (getCodeBase(),"pic/e"+picx+"n"+picy+"s4.jpg");
        repaint();
}
```

```
        public void MoveUpLeft()
        {
                picx = picx -2400;
                picy = picy + 2400;
                i = getImage (getCodeBase(),"pic/e"+picx+"n"+picy+"s4.jpg");
                repaint();
        }
        public void MoveDownLeft()
        {
                picx = picx - 2400;
                picy = picy - 2400;
                i = getImage (getCodeBase(),"pic/e"+picx+"n"+picy+"s4.jpg");
                repaint();
        }
        public void MoveUpRight()
        {
                picx = picx + 2400;
                picy = picy + 2400;
                i = getImage (getCodeBase(),"pic/e"+picx+"n"+picy+"s4.jpg");
                repaint();
        }
        public void MoveDownRight()
        {
                picx = picx + 2400;
                picy = picy - 2400;
                i = getImage (getCodeBase(),"pic/e"+picx+"n"+picy+"s4.jpg");
                repaint();
        }
}
```